Observe. Trace. Intercept. Verify.

Updated from the supplied four-page ReefTrace Rig Guide and the current repository. This is the physical build guide; it does not replace the software-only teammate assignments.

Tideward films one floating red token in a small freshwater tank, compares which declared entry region fits its motion, then tests whether a collector placed upstream reduces entry into a marked protected area.

Current status: the first physical trial has not run.

The app, synthetic rehearsals and historical NOAA studies exist. The wet rig still needs assembly, actual webcam verification, measured calibration and new independent releases. A software pass is not a tank result.

The entire rig has three connections

1 USB video

C920s webcam
attached USB-A cable
to the laptop

2 Pump power

Original pump cord
to grounded GFCI power
manual on / off

3 Water loop

Pump to fitted tube
to submerged outlet
back into the same bin

The pump makes circulation. It does not suck up plastic. The mesh pocket catches the token, and a person retrieves it. The laptop does vision, motion fitting, source support, layout comparison and evidence storage.

Start in this order

Pages	Do this	Pass before moving on
2-3	Audit inventory and acquire missing essentials	Actual part in hand; fit checked
4-7	Connect, assemble and measure	Stable, dry camera; contained circulation
8-9	Commission each physical layout	Mapping, drift error and support gates pass
10-11	Run new trials and explain the evidence	Frozen plan; honest complete denominators

No ESP32-CAM, MIPI module, Arduino pump wiring, relay, servo, 3D printing or decorative coral is needed for this build.

Use what you actually have.

Status is based on the team-provided inventory and purchase screenshots. A reported purchase is not a physical fit test. Tick the first two columns before another shopping run.

Reported status	Item	Required action
REPORTED BOUGHT	Pond Boss 80 GPH corded pump	Read its actual model/label. Confirm 99034, original cord, two supplied outlet adapters and undamaged intake.
REPORTED BOUGHT	Sterilite 6 qt bin	Measure inside length/width and safe water depth. Check rigidity, leaks, pump submersion, freeboard and workspace.
REPORTED BOUGHT	Top Fin mesh breeder	Check actual fit and surface interception. Secure its mesh as a compact pocket if the full frame obstructs the bin.
OWNED	Black tape	Use for position marks and dry cable strain relief. It does not replace a hose clamp or a waterproof connection.
EXPECTED / CHECK	Laptop + charger	Confirm working repo, camera permissions, correct data USB port and dry placement.
CHOSEN / NOT VERIFIED	Logitech C920s USB webcam	Confirm possession. If missing, borrow/buy the selected camera before physical commissioning.
CHECK / ACQUIRE	Mount, tubing, clamps, tokens, ruler and tray	Exact quantities and acceptance checks are on page 3.

Keep the listed electronics in the kit

Do not connect them to this rig: UNO R3 x2; HW-233 slider; ULN2003 board (no stepper listed); solder PCB, breadboards and jumper wires; 3 W speaker; HC-SR04 x2; LCD1602; Elegoo MB V2; QA052 car shield; Waveshare display; UNO proto shield; QA007 ESP32; OLED; ESP32-CAM; OV5670 MIPI camera; SG90; DHT11.

They solve no required function in this USB-camera/manual-pump plan. The bare ESP32-CAM lacks its complete programming path; the MIPI camera does not plug into USB. The SG90 and the ultrasonic/DHT modules are not water sensors for this experiment.

Coral and 3D printing: no purchase or print required.

The protected area is a measured polygon in the water-plane map. Decorative coral adds no measurement and can obscure the token; red coral can confuse red tracking. A rigid borrowed camera stand is sufficient.

Acquire only the missing items.

Every function below is required. Buy or borrow only after the inventory audit; do not repurchase the pump, bin or mesh merely because they appear in this guide. Product links are clickable.

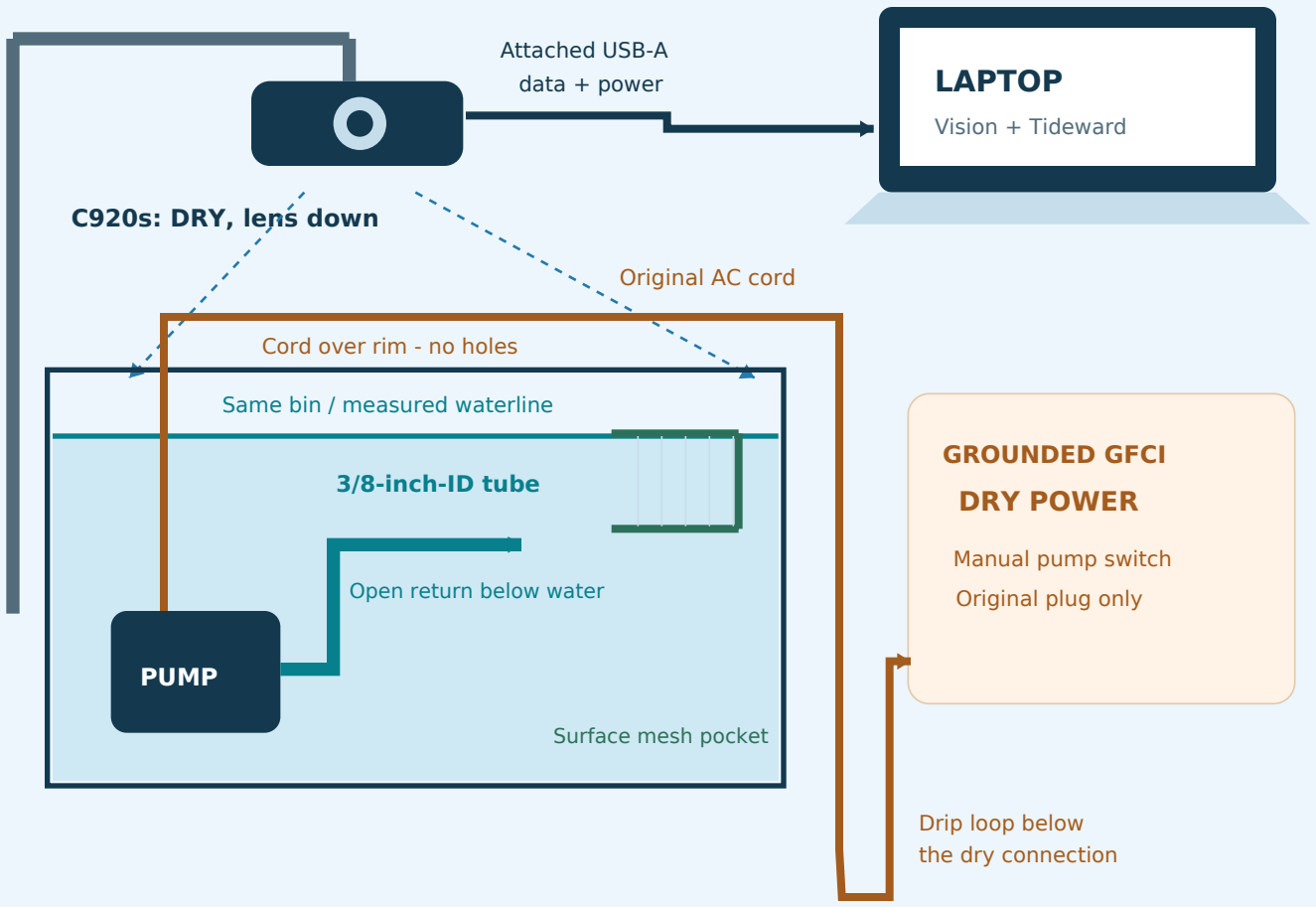
Qty	Required item	Exact acceptance check
1	Logitech C920s	External USB webcam selected in the app. Its attached USB-A cable carries video and power. No separate camera supply.
1	Rigid overhead camera mount	Desk clamp or stable boom/tripod with compatible 1/4-inch camera fitting. Lens points down; no sag; whole water plane visible.
0 or 1	USB-A to host-port DATA adapter	Required only if the laptop has no usable USB-A port. Test camera video through it; a charging-only cable/adapter is unsuitable.
About 3 ft / 0.9 m	Flexible vinyl tube, 3/8-inch INSIDE diameter	Fit the actual supplied 3/8-inch barb first. Inside diameter is 9.525 mm; outside diameter varies. Cut only after routing.
1 + 2	Small hose clamp + two outlet anchors	Clamp fits the tube outer diameter at the barb without crushing it. Two clips/zip ties secure the return end to a rigid support.
1 set	Mesh pocket and secure rim fasteners	Use reported-bought breeder mesh. Keep mouth open at surface, no under/side escape; no sharp edges or loose pieces.
3 identical	Reusable red floating tokens	Same material/size; test visibility and float stability. One token in water at a time; recover and count every token.
1 set	Ruler, marker tabs, tray, towels, steady light	Measure in cm. At least 4 mapping marks + 3 distinct check marks at water height. Use existing room light if stable.
1 location	Grounded, GFCI-protected dry power	Original pump plug only. Dry elevated connections and drip loop; venue-approved supply. No breadboard/mains modification.

If the supplied 3/8-inch barb is actually missing, use the manufacturer replacement 3/8-inch adapter, SKU 53458 after confirming pump compatibility. Do not stretch 3/8-inch tubing over the 1/2-inch barb.

The 6 qt bin is usable only if the complete assembly passes the fit/depth test. If it fails, obtain one larger stable watertight bin before proceeding; this is a fit failure, not a new design branch.

This is the complete wiring.

Physical connection diagram - not to scale. BLACK/BLUE = USB data, ORANGE = original AC cord, TEAL = water tube. No signal or power wire connects the laptop to the pump.



Only these two electrical plugs are used.

Webcam USB into laptop; pump original plug into grounded GFCI-protected power. The tube contains water, not wires. The pump is manually controlled. Software Stop does not disconnect pump power.

Place the laptop and all plugs outside the splash area. Use fresh water, keep the pump fully submerged and keep its intake clear. Unplug before wet adjustments or reaching into the tank. Follow the actual pump label and manufacturer manual.

Build the wet loop first.

Do this with the pump unplugged and the laptop away from the filling area. Water circulates entirely within the one bin; there is no external reservoir or suction hose.

1 Place and measure the bin

Put the bin on a rigid level surface inside a spill tray. Add fresh water to a depth that fully covers the actual pump, leaves working space and safe freeboard, and stays contained when the pump runs. Mark that waterline. Measure actual internal length L and width W at that plane. Do not use the old guide's fixed 7-8 cm as a guarantee.

2 Fit the correct barb and tubing

Check the physical model number. For 99034, select its supplied 3/8-inch-ID tubing adapter; fit it to the pump outlet using the manufacturer instructions. Slide 3/8-inch-ID tube onto that barb, then secure the fitted joint with a correctly sized hose clamp. A dry gentle pull must not detach it. Black tape is a position marker, not the retaining joint.



3/8 inch = INSIDE diameter = 9.525 mm. Tube outer diameter is different.

3 Secure the return and keep the intake clear

Route the tube inside the bin, without kinks. Secure its open return end at two points, submerged and aimed consistently across the surface region. Keep the pump intake away from the tokens and collector, unobstructed and protected by its original screen. No nozzle aimed into the air. Measure the necessary route before cutting the approximately 3 ft supply.

4 Make the surface collector

Use the bought breeder mesh as a pocket whose open mouth faces the observed flow. Secure the rim; make the pocket span the token's surface level. Use the full frame only after it fits. If not, secure its mesh as a smaller pocket with existing clips and a smooth support. Measure the actual mouth and footprint. No preset 6 x 14 cm cutting pattern is assumed.

Before calibration, test one reusable token: it must float, remain visible and be retained without escaping under, over or around the pocket. Recover it manually. This development check is not a final prevention trial.

Fix the camera. Lock the setup.

The webcam stays above the water. Use the selected external USB camera, not an improvised immersed camera or the incomplete ESP32-CAM.

5 Fix the overhead webcam

Fasten the C920s to a rigid overhead support using its compatible 1/4-inch mount or secure camera clamp. Point the lens down. Frame every source region, the complete protected zone, both planned collector locations and all calibration marks. The arm must not move when a cable is touched. Strain-relieve the USB lead on the dry support.

6 Connect USB and open the correct camera

Connect the webcam's attached USB-A lead to the laptop, using a data-capable host adapter only if the port needs one. No separate webcam power supply is used. In Tideward's Camera wizard, enter the external camera device index, click **Start this webcam**, verify the actual tank preview and choose **Use current session**. Grant the OS camera permission privately if asked.

7 Test the pump and tracking

With the pump fully submerged and the outlet secured, start at its minimum flow setting using dry hands and the original grounded/GFCI power. Observe circulation without overflow, spray or a detached hose. Adjust the development setup only while safely unplugged. Drop one red token, check that tracking follows it, then retrieve it. Do not expect every release to enter the pocket.

Visibility pass

Use steady diffuse light from the side, a matte background and a clean water surface. Remove other red objects. Eliminate the bright reflection before trusting the tracker. Verify the same token is detected across the intended observation region; two red blobs or a gap do not become a continuous trajectory.

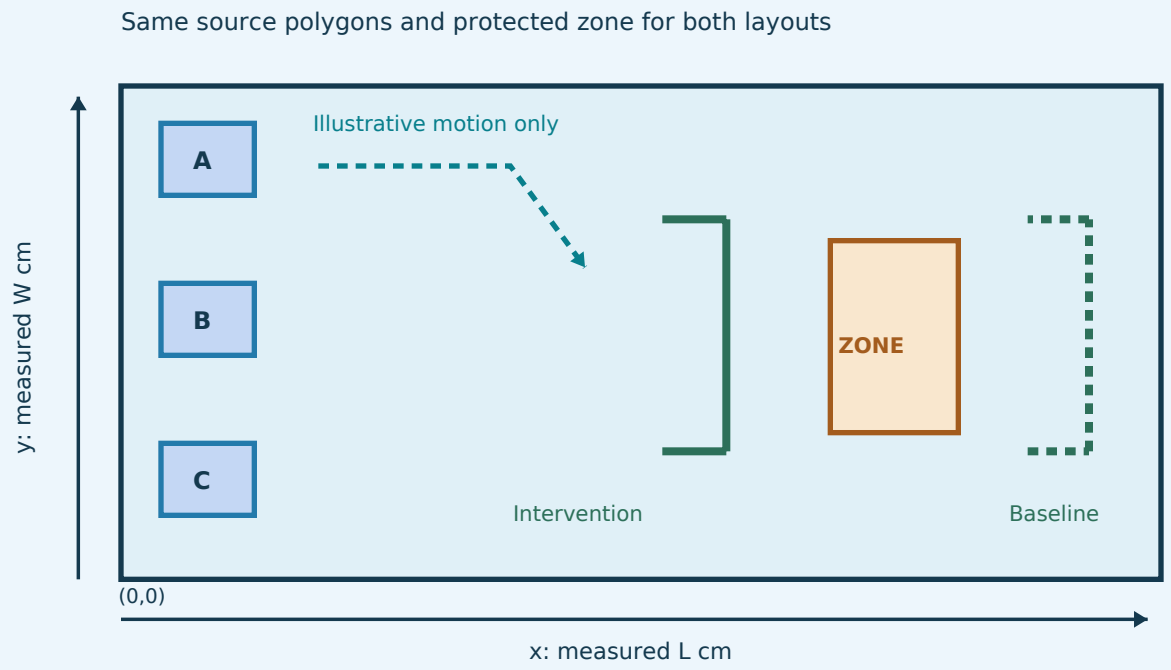
8 Mark and record the frozen configuration

Mark the bin, camera/support, waterline, pump location/dial, return position/direction and each collector location. Save a private setup photo and a measured sketch. Record the camera session, token class and lighting. Do not move or refill the rig after measuring it without declaring the disturbance and doing the applicable revalidation.

A pump label such as 80 GPH is a device rating, not the local surface velocity. Tideward must measure the token drift in this exact setup. The manual requires full submersion; a 6 qt bin alone does not prove adequate operating depth.

Draw the map you can measure.

SCHEMATIC ONLY - not a scale drawing, pump calibration or recommended set of coordinates. L, W and every polygon vertex must come from your physical rig in centimetres.



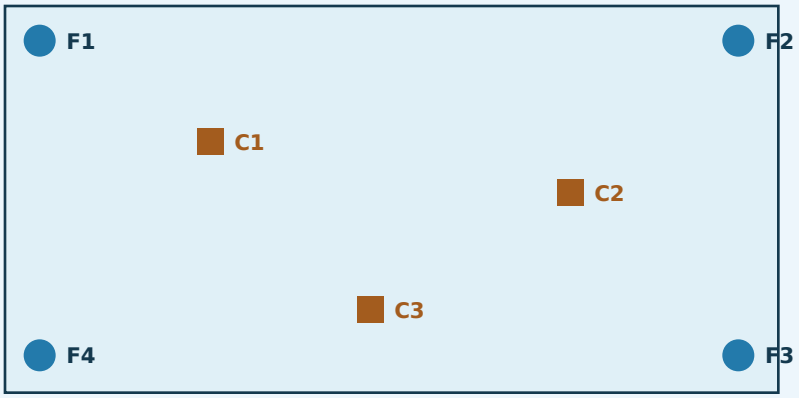
The solid and dashed collectors are **two alternative positions of one pocket**; do not install both at once. In this sketch intervention is before the zone, baseline after it. Actual circulation can curve or reverse: choose positions from development measurements, not the arrow in the picture.

Record before fitting	Your physical measurement
Water-plane bounds	L = _____ cm; W = _____ cm; water depth = _____ cm
Source regions A, B, C	Each measured polygon: [(x1,y1), (x2,y2), ...]
Protected and observation regions	Two separately declared polygons; keep the zone clear and visible
Collector in each layout	Measured footprint, mouth/retention region and fixture marks

Use the same origin, axes and units everywhere. Keep protected zone, source polygons and collector polygons separate; the comparison rejects touching/overlapping forbidden regions. Decorative coral is not the protected-zone measurement.

Four fit points. Three new checks.

Open Camera > Field commissioning. The mapping converts pixel locations into positions on the WATER surface. Bin-floor measurements through water are the wrong plane.



F1-F4 = FIT

C1-C3 = CHECK

All seven positions measured at water height

Place marks firmly at the surface height. Remove temporary markers before drift.

1. Establish a stable reference at water height, for example fixed rim-supported tabs or a temporary supported calibration card. Use at least seven well-spread visible marks. Measure each mark's x and y from the chosen origin with a ruler. Do not assume the printed drawing has the right scale.

2. In the wizard choose centimetres. Enter at least four fit correspondences: click their pixels in the preview and type their measured water-plane coordinates. Spread points around the usable area; avoid collinear marks.

3. Enter at least three **different** check marks, excluded from fitting. Declare the maximum allowed check-point error before fitting, using development measurement precision. Click **Fit and validate mapping**. The stored check errors must pass that declared tolerance.

4. If it fails, fix point identity, height, units, glare or mount and repeat with a new mapping. Do not increase the tolerance to turn failure into approval. Save calibration ID, actual errors and camera session. Remove the calibration aid without moving the camera, outlet or waterline.

When calibration becomes stale

A moved camera, changed image geometry or water height changes the map. Correct the setup and recalibrate as required; do not reuse old centimetres. Camera timestamps are host receipt times, not hardware exposure timestamps.

Measure both layouts separately.

A spatial drift field is learned from token motion in this tank. A NOAA trajectory, simulated field or copied baseline field is not a measured intervention field.

Do this for each layout	Existing software action / pass condition
1 Declare development settings	Fix waterline, token class, geometry, support radius, cell size, error limits and intended horizon before evaluating validation runs.
2 Record training drift	Use Record training run for at least two separate one-token releases. Spread development releases through the region the app must predict.
3 Record new validation drift	Use Record validation run for separate releases. No shared recording IDs or frames; do not split a single run into pretend independent releases.
4 Fit and inspect	Fit recorded flow. Inspect recorded-flow vectors (cm/s) and free-running paths (cm). Gaps and unsupported regions stay visible.
5 Admit the measured profile	Enter measured source/observation/pocket polygons. Held-out coverage must reach the existing 90% gate and error must pass the predeclared positive limit.
6 Save lineage	Record session, calibration, recording, flow and profile IDs plus hashes. Repeat with fresh, disjoint recordings for the other collector position.

Validation is a check, not training. If its coverage or error fails, the current rig is not admitted. Add development measurements or repair the setup, then freeze a new configuration and collect new validation. Do not widen a support radius or error limit just to pass.

After a bump, retrieval, refill or layout change

Declare the disturbance, restore all marks and follow the Rig state checks. Use **Record drift and revalidate** when required. Restoration/revalidation is not permission to reuse a profile for a genuinely changed geometry; changed layouts require their own admitted measured profiles.

For the initial layout, record a fresh source observation under the admitted profile. Complete physical attestations, click **Check all steps**, and freeze the field receipt only when every check passes. READY records verified artifacts plus operator statements; it is not independent certification.

Keep source truth out of inference.

Finish development and admission before final evaluation releases. Start with one token per release. Record real results even when they disagree with the model.

A First real source observation

Restore the measured layout and satisfy its rig-state gates. The evaluator chooses A, B or C and records that release answer privately. Use **Record new source case** for a fresh recording under the admitted profile. Save its source report before revealing the answer to the person judging it. Known release labels are evaluator truth, never estimator inputs.

A **Ranked** report means the declared candidates are distinguishable under the frozen model. **Unknown** means insufficient/ambiguous evidence, including many already-retained endpoints. **Invalid** means a gate failed. Relative source support is not a calibrated polluter probability.

B Freeze the interception pilot

Select the real source case and two compatible, separately admitted MEASURED_TANK profiles. Keep source polygons, token, bounds and protected zone identical. Freeze the horizon and a pilot quota of **trials_per_source = 1**: three sources x two layouts = **six fresh releases**. If the UI keeps its default five-per-cell quota, it is a 30-release plan; do not call six a completed 30.

Use only a horizon supported by the measured setup, up to the current 20 s workflow limit. Freeze before any final release. Follow the saved Next trial order exactly; layout changes require restoration/revalidation. Do not recompute the anchored source case mid-plan because that archives the comparison.

C Record what happened first

Observed outcome	How to record it
Zone entry	Record first entry time. Later collection does not undo that arrival.
Intercepted before entry	Token must remain captured through the full frozen horizon; account for it.
No entry by horizon	Observe the entire horizon and account for the token.
Lost / obscured / unaccounted	Unresolved. Keep it in the denominator; never count it as prevention.

Begin each linked recording after plan freeze and use it for one release only. Save exact trial key/body, layout, evaluator source, event time, horizon completion and token accounting. On a network timeout retry the same key/body; do not release a second token to repair a request.

Show what the rig actually proved.

Current physical results remain zero until the real rig is assembled, commissioned and its new releases are recorded. This guide is an execution plan, not a completed experiment.

What to show in Tideward

Start: from the reeftrace repository use `.venv/bin/python scripts/dev.py`, then open **http://127.0.0.1:3000/**. Use the private local operator entry; credentials stay off slides and screenshots. Do not restart an already running service merely to follow this guide.

Investigations: observed path and conditional source support. **Prevention:** simulated first-zone-entry predictions versus separately labelled actual outcomes. Missing/unresolved slots widen descriptive bounds; bounds spanning zero do not establish improvement. **Camera:** calibration, drift and rig lineage. **Data lab:** historical NOAA motion research, separate from tank evidence.

Chalk: ask it to explain the exact saved source/comparison/evaluation and cite tool evidence; it does not measure water, edit outcomes or operate the pump. **Photon:** preserve the current paired line and reply configuration. This rig build sends no new test message and changes no provider settings. Software Stop stops software work, not AC pump power.

Four demo jobs - without reassigning software lanes

Job	Owner and boundary
Rig and safe water loop	Jose leads the physical build; physical assistant is unassigned.
Camera and commissioning	Jose, assisted only by a person explicitly assigned later.
Release truth / outcome log	Assign a separate evaluator for the pilot; hide source truth until prediction is saved. Sequence jobs if staffing is limited.
Dashboard / evidence narration	Jose coordinates; Megana keeps her software demo/a11y/E2E lane and Aarush keeps export/validator software work unless reassigned.

Final go / no-go

GO only with a stable mount, correct tube/clamp fit, contained submerged circulation, one visible recovered token, approved water-plane mapping, independent admitted drift for both layouts, fresh source case and a frozen trial plan. Otherwise report the exact blocker and present only the labelled software rehearsal.

Verified references and corrections

Pond Boss 99034 product | actual pump manual | 3/8-inch replacement adapter
C920s product | C920s USB/mount setup guide | Top Fin mesh breeder
Repository / current software-only handoffs | Field commissioning | Integrated demo

Links checked 17 Oct 2026. The original guide's fixed depth/hose/pocket sizes, tape-as-clamp, cut-foam batch and claimed prices are superseded by measured fits, a retaining clamp and reusable tokens. No physical run, camera possession or product availability is inferred from a screenshot.